Phenomenological T_m models — equations

Shared two-state core

$$T_m(^\circ C) = \frac{1000 \sum_k g_k \Delta H_{NN,k}}{\Delta S_{eff}} - 273.15, \quad g_k = e^{-\acute{d}_k/\lambda}$$

Here $\acute{d}_k$ is the mean distance of bond k to the other bonds, N is the number of bonds (2 per binding site), and every ΔS_{eff} also includes the bimolecular term $+R\ln(C_T/2) \approx -39.6$ cal/mol·K.

Family A — occupancy-weighted

A1 — mean: $\Delta S_{eff} = \sum_k g_k (\Delta S_{NN,k} + \Delta S_{Bond}) + base$

A2 — N-scaled ΔS_{Bond} (best): $\Delta S_{eff} = \sum_k g_k \Delta S_{NN,k} + N \Delta S_{Bond} + base$

A3 — g on ΔH only: $\Delta S_{eff} = \sum_k \Delta S_{NN,k} + N \Delta S_{Bond} + base$ (the weight g_k is applied only in the ΔH sum)

A4 — nearest-neighbor g: as A1 with $g_k = e^{-d_{min,k}/\lambda}$ ($d_{min,k}$ = distance to the single closest bond)

A5 — per-type ΔS_{Bond}: $\Delta S_{eff} = \sum_k g_k (\Delta S_{NN,k} + \Delta S^{type}_{Bond}) + base$

A6 — split λ_H / λ_S: ΔH uses $g(\lambda_H)$ and ΔS uses $g(\lambda_S)$

A7 — occupancy + rotational: A2 with $-k_{rot} R \ln\left(1 + \sum_k (z_k - \acute{z})^2\right)$ added to ΔS_{eff}

Family B — elastic / mechanical-coupling

B1 — g = 1 (no weighting): $\Delta S_{eff} = \sum_k \Delta S_{NN,k} + N \Delta S_{Bond} + base$ with $g_k \equiv 1$

B2 — g = 1 + rotational: B1 with $-k_{rot} R \ln\left(1 + \sum_k (z_k - \acute{z})^2\right)$

B3 — g-weighted + WLC tethering: $\Delta S_{eff} = \sum_k g_k (\Delta S_{NN,k} + \Delta S^{type}_{teth}) + base$

B4 — WLC tethering only: $\Delta S_{eff} = \sum_k (\Delta S_{NN,k} + \Delta S^{type}_{teth}) + base$

Mechanical terms

Rotational (elastic Green's-function):

$$\Delta S_{rot} = -k_{rot} R \ln\left(1 + \sum_k \left(z_k - \acute{z}\right)^2\right)$$

WLC tethering entropy (derived; $\ell_p = 1$ nm, rise 0.63 nm/nt):

$$\Delta S_{teth}^{type} = R \ln \frac{V_2}{V_1^2 C_0}, V = 4/3\,\pi \langle r^2 \rangle^{3/2}$$

The 6nt bond is asymmetric (8 nt on one tile, 7 nt on the other):

$$\Delta S_{teth}^{6} = R \ln \frac{V(1.5)}{V(8)V(7)C_0}$$